Conversation State and Memory

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected state and retention record.

Explain the policy, trace corrections, audit a shortcut and design an expiry test.

Keep one habit: Keep provenance and lifetime beside every value; distinguish old history from current state.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

User Need and AI Contribution

Model Memory vs Retrieval

Hallucination Failure Types

W1 · Explain state and lifetime

> Separate active fields, derived summaries and retained history.

Your evidence record

- Session-only policy: one current selection; correction replaces it; derived summaries must refresh/invalidate; minimal correction audit lasts only until exit; exit clears state, summary and audit; new session with no input asks. Room practice is A→B with stale summary A. Book example is Cedar→Maple→cleared. These application records do not establish permanent model learning or deletion from external services.

- Reference for this exercise: open this week's student lesson, "Conversation State and Memory", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the room correction

> Invalidate a stale summary after replacing A with B.

Your evidence record

- Session-only policy: one current selection; correction replaces it; derived summaries must refresh/invalidate; minimal correction audit lasts only until exit; exit clears state, summary and audit; new session with no input asks. Room practice is A→B with stale summary A. Book example is Cedar→Maple→cleared. These application records do not establish permanent model learning or deletion from external services.

- Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit a memory guarantee

> The toy session policy does not establish permanent model learning.

Your evidence record

- Session-only policy: one current selection; correction replaces it; derived summaries must refresh/invalidate; minimal correction audit lasts only until exit; exit clears state, summary and audit; new session with no input asks. Room practice is A→B with stale summary A. Book example is Cedar→Maple→cleared. These application records do not establish permanent model learning or deletion from external services.

- A report omits this qualification: Remembered **state**ments are not automatically current or verified; retention should follow the **state**d purpose and policy.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing state test

> Supply events, expected values and a correction/reset boundary.

Your evidence record

- Session-only policy: one current selection; correction replaces it; derived summaries must refresh/invalidate; minimal correction audit lasts only until exit; exit clears state, summary and audit; new session with no input asks. Room practice is A→B with stale summary A. Book example is Cedar→Maple→cleared. These application records do not establish permanent model learning or deletion from external services.

- Reference for this exercise: open this week’s student lesson, “Conversation State and Memory”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Trace the new lab-selection session independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Apply the same explicit session-only policy to the supplied lab selection: one active value, correction replaces it, exit clears active state and session audit, and a new session without input has no selection. Trace the stages from the given events; no persistent store or real service is supplied.

A session stores lab L1 then corrects to L2; session ends. A new session has no room **input**. **State** the active values at each stage.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Store minimal **session state** and make **persistent memory** optional, visible, and erasable.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Conversation **state** and retention. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Conversation **state** and retention. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

The conversation **state** is topic=plants, level=Grade 8. The user says 'Keep the topic, explain for Grade 6'. Write the updated **state** and a reset behaviour.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

State stores topic=plants and reading_level=beginner. User requests advanced explanation.

What changes?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A learner corrects their project topic from oceans to forests. The next reply still uses oceans. What **state** should change, and what should not be saved unnecessarily?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 20 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

The conversation **state** is topic=plants, level=Grade 8. The user says 'Keep the topic, explain for Grade 6'. Write the updated **state** and a reset behaviour.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A session stores lab L1 then corrects to L2; session ends. A new session has no room **input**.
State the active values at each stage.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

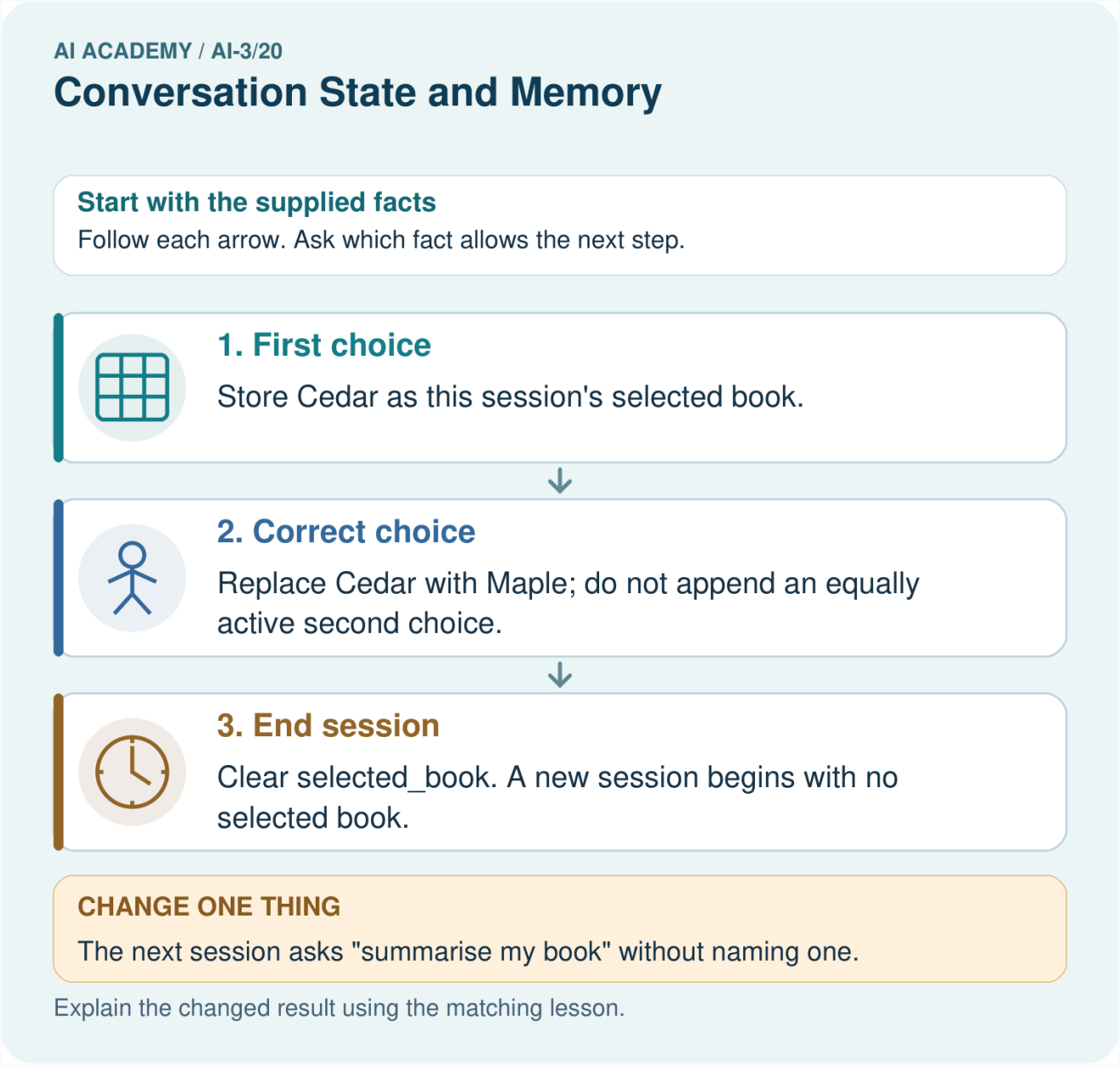
> Keep the explanation and the support you used.

Use this when helpful

Keep a reading conversation consistent when the learner changes the selected book.

In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Keep a reading conversation consistent when the learner changes the selected book.
- Which labelled step matches this detail? Clear selected_book. A new session begins with no selected book.
- 1. First choice

- 2. Correct choice
- 3. End session
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: End session Clear selected_book. A new session begins with no selected book. Read the picture in order. Clear selected_book.

  A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The next session asks “summarise my book” without naming one. Which conclusion is supported by the supplied facts?
- 1. Ask which book the learner wants.

- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Ask which book the learner wants. The declared policy clears **session state**, so yesterday's choice is unavailable.

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

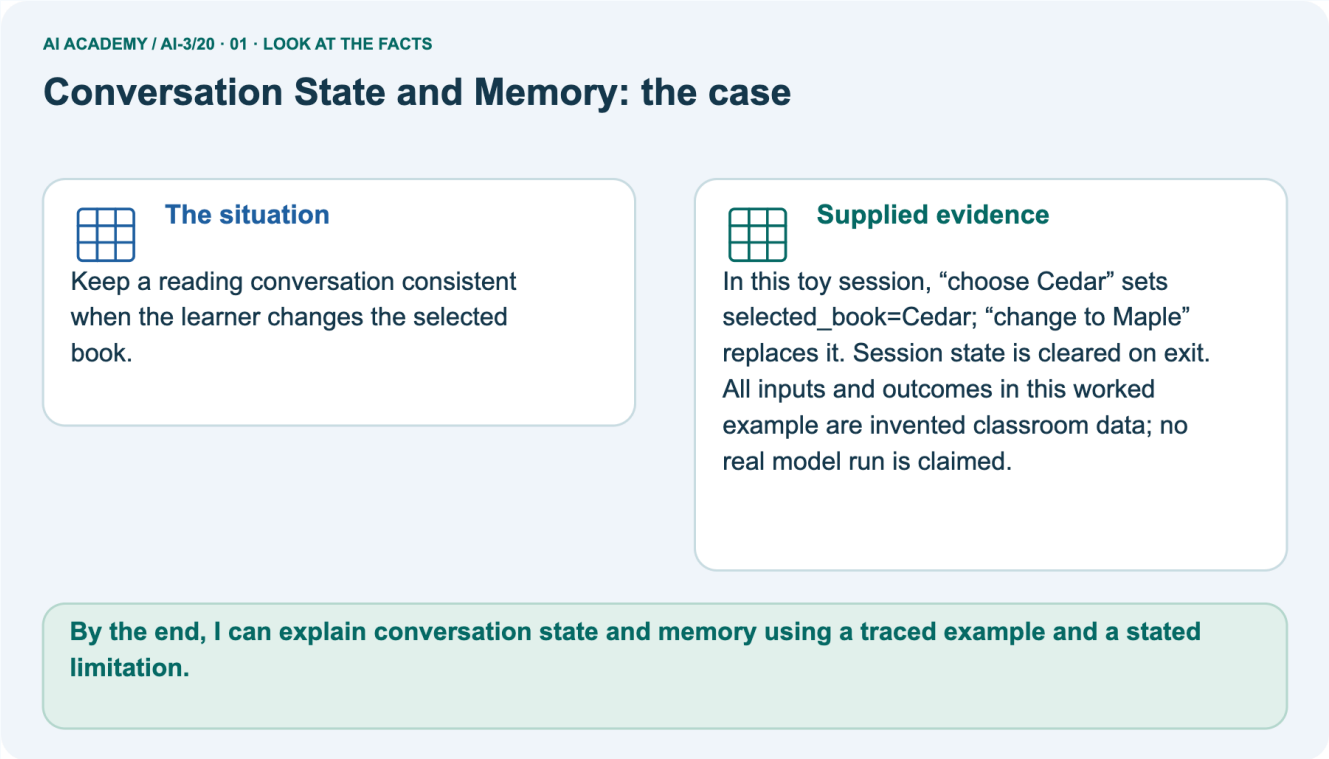
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Conversation State and Memory: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

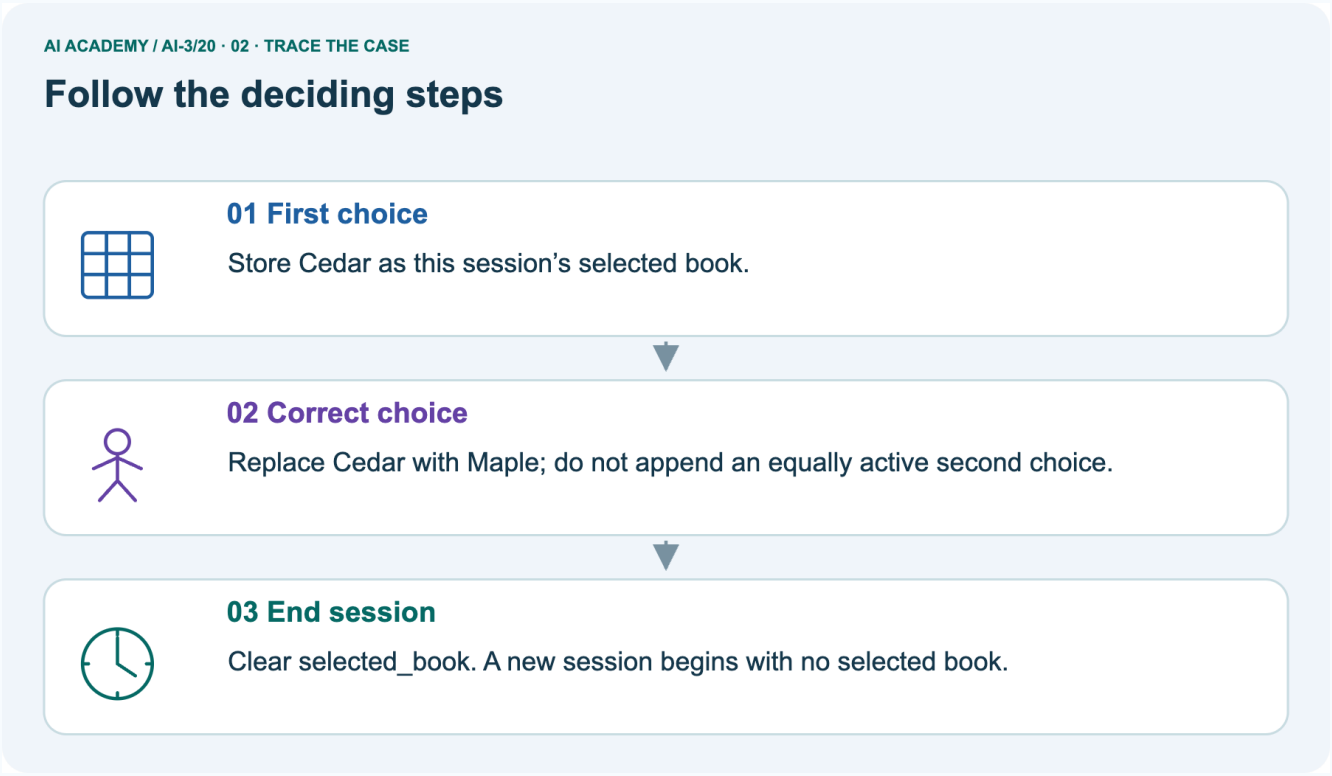
Keep a reading conversation consistent when the learner changes the selected book. Supplied facts: In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

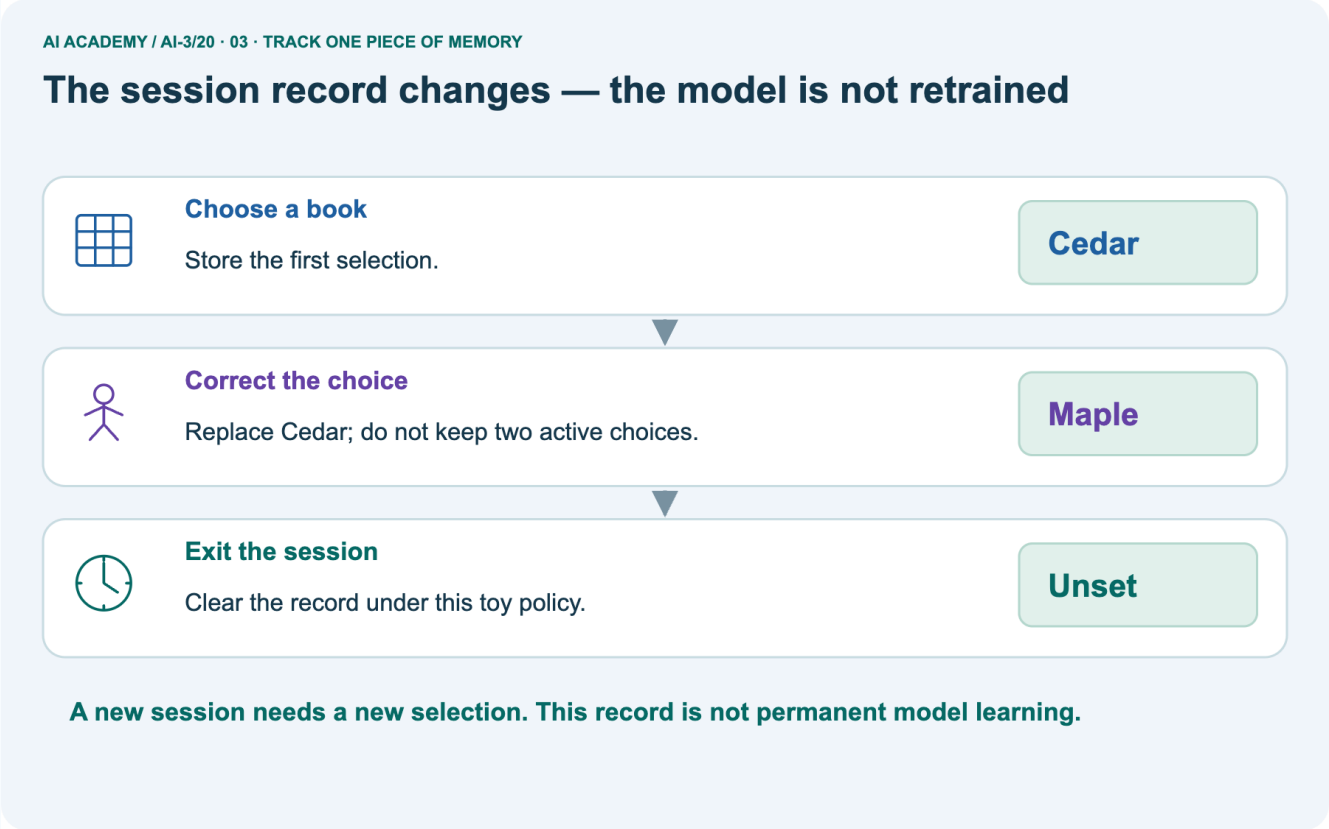
First choice: Store Cedar as this session's selected book. → Correct choice: Replace Cedar with Maple; do not append an equally active second choice. → End session: Clear selected_book.

A new session begins with no selected book.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. The session record changes — the model is not retrained



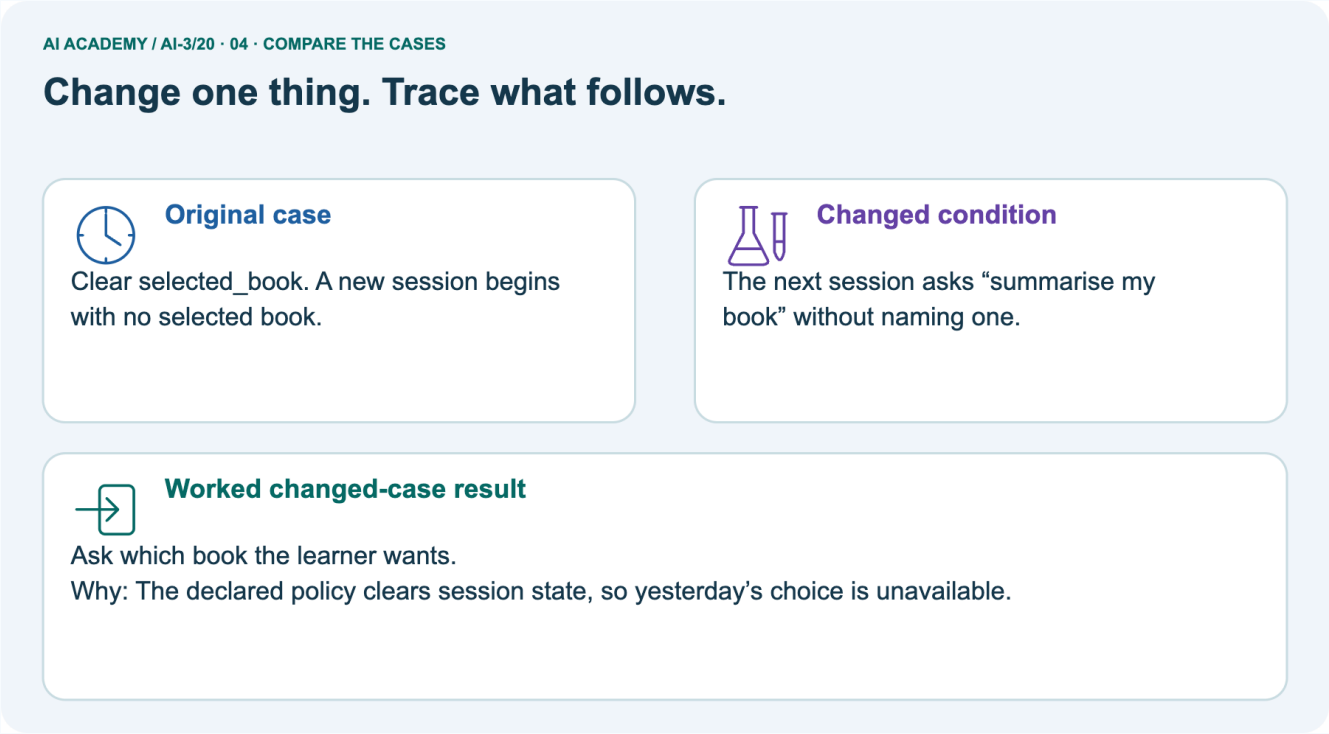
Follow the labelled picture with the matching explanation. Open the vector diagram

The selected_book **session state** is Cedar after the first choice, Maple after correction, and unset after exit. This is an application record, not proof of permanent learning.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

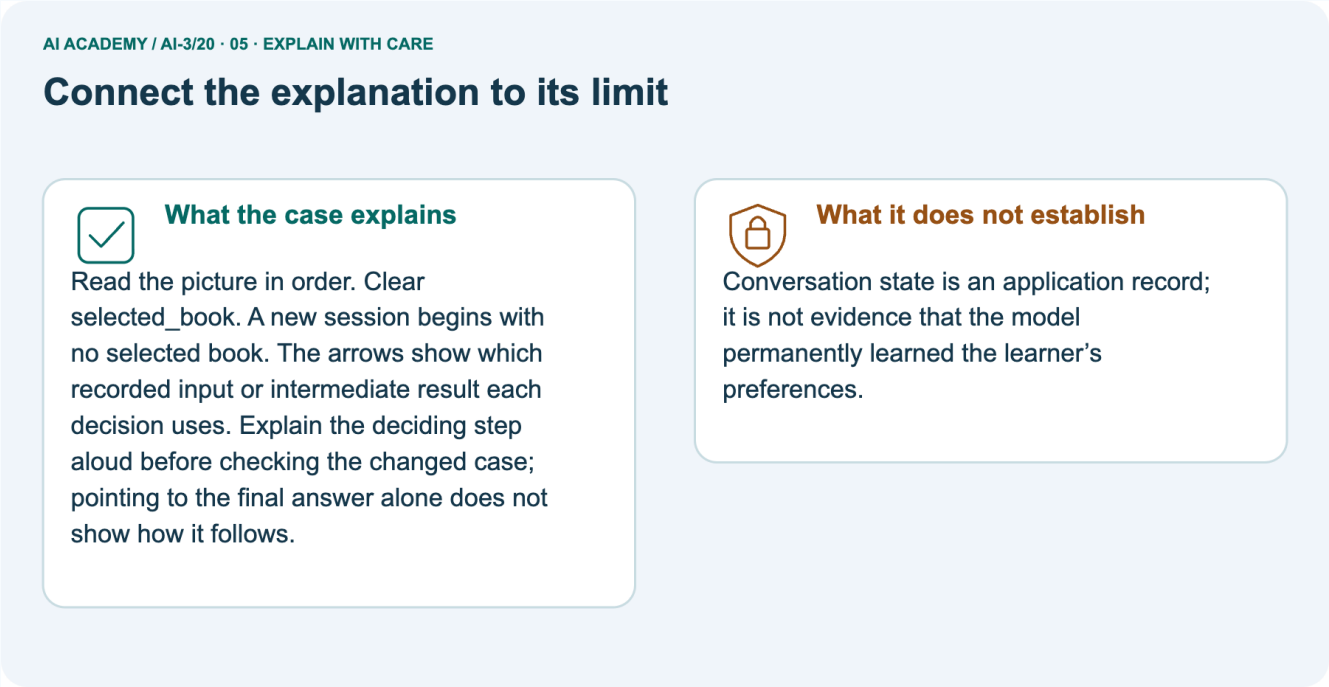
Original case: Clear selected_book. A new session begins with no selected book. Changed condition: The next session asks “summarise my book” without naming one. Expected result: Ask which book the learner wants.

Reason: The declared policy clears **session state**, so yesterday’s choice is unavailable.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Clear selected_book. A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

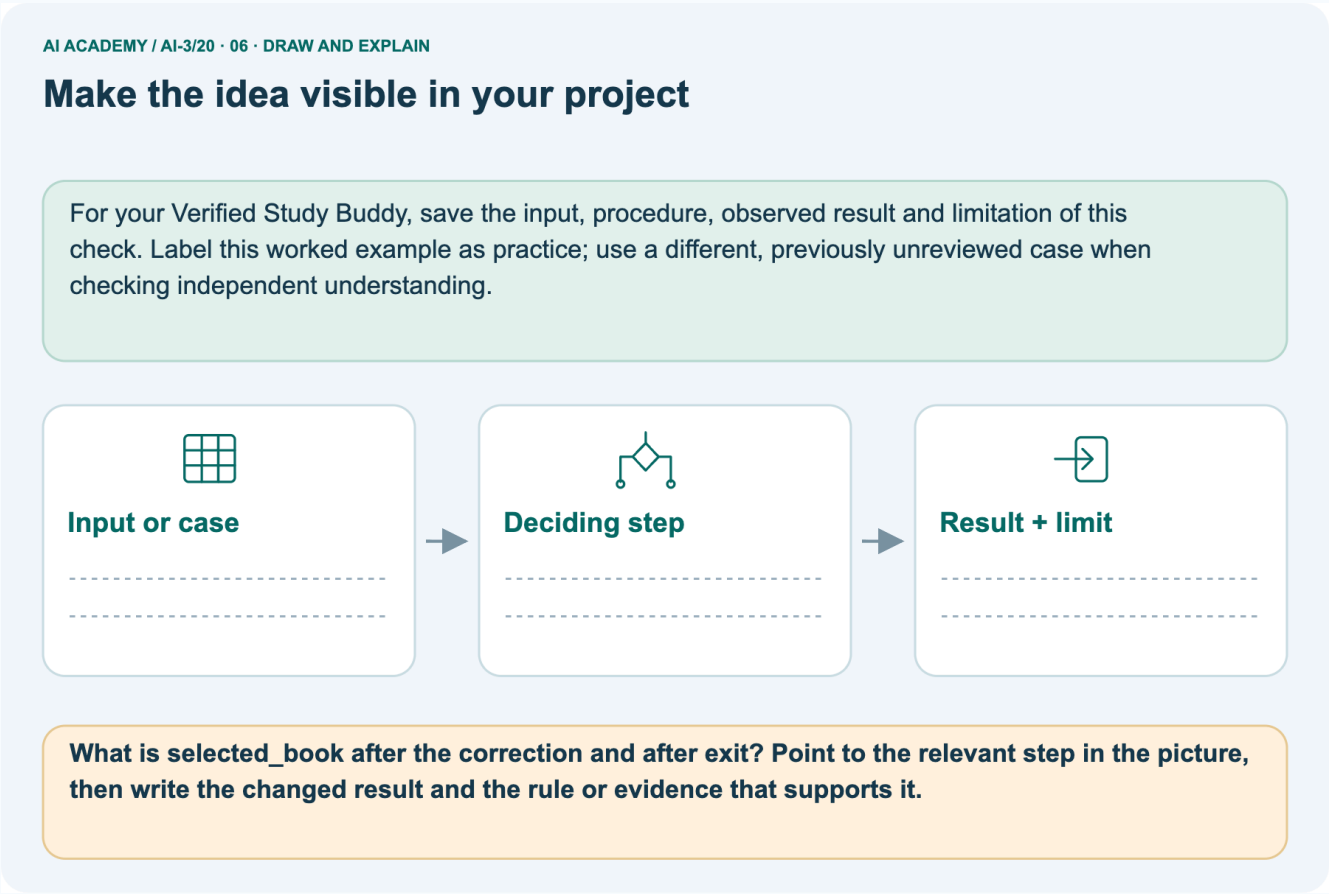
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Conversation **state** is an application record; it is not **evidence** that the model permanently learned the learner's preferences.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Conversation State and Memory: the case

02. Follow the deciding steps

03. The session record changes — the model is not retrained

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.